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SINCE 2019

CUSTOMER CHURN PREDICTION AND ANALYSIS

Course: Data Science

Submitted By:

TEJASHWINI RAMESH HUBBALLI

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Customer Churn Prediction and Analysis

1. Abstract

Customer churn is one of the most critical challenges faced by subscription-based businesses such as streaming platforms. When customers stop using a service, it directly affects revenue, profitability, and long-term business growth. Therefore, identifying customers who are likely to churn is essential for companies to take proactive actions and improve customer retention.

This project focuses on analyzing customer behavior and predicting churn using data analytics and machine learning techniques. The dataset used in this project contains customer information such as subscription type, watch hours, device usage, region, and login activity. Various data preprocessing and exploratory data analysis techniques were applied to understand patterns and relationships within the dataset.

A Logistic Regression machine learning model was implemented to predict whether a customer is likely to churn or remain active. The model was evaluated using metrics such as accuracy, confusion matrix, and classification report.

In addition to predictive modeling, an interactive Power BI dashboard was developed to visualize key insights such as churn rate, device usage, customer engagement, and subscription distribution. The results of this analysis help identify important factors influencing churn and provide business recommendations to improve customer retention.

2. Introduction

Customer churn refers to the situation where customers stop using a company's product or service over a period of time. In industries such as streaming services, telecommunications, and subscription-based platforms, customer churn is a major concern because losing customers directly impacts business revenue.

Streaming platforms such as Netflix, Amazon Prime, and Disney+ operate in highly competitive markets where users have multiple alternatives available. As a result, customers can easily switch between services if they are not satisfied with the content, pricing, or overall experience.

Predicting customer churn allows businesses to identify customers who are at risk of leaving the platform. By analyzing historical customer data such as watch behavior, device usage, subscription plans, and login activity, companies can detect patterns that indicate potential churn.

Data analytics and machine learning techniques play a crucial role in churn prediction. These technologies allow businesses to analyze large volumes of customer data, identify trends, and develop predictive models that estimate the probability of churn.

This project aims to build a customer churn prediction system using SQL, Python, and Power BI. The project includes data preprocessing, exploratory data analysis, machine learning modeling, and dashboard visualization to provide meaningful business insights.

3. Project Objectives

The main objectives of this project are:

- To analyze customer behaviour using data analytics techniques.
- To identify key factors that influence customer churn.
- To perform exploratory data analysis to understand engagement patterns.
- To build a machine learning model to predict customer churn.
- To evaluate the performance of the predictive model.
- To develop an interactive dashboard for visualizing churn insights.
- To provide business recommendations for improving customer retention.

4. Tools and Technologies

Several tools and technologies were used to successfully implement this project.

Python

Python was used as the primary programming language for data analysis and machine learning. It provides powerful libraries for data manipulation, visualization, and predictive modeling.

SQL

SQL was used to store the dataset and perform analytical queries. It helped in calculating important metrics such as total customers, churn rate, subscription distribution, and device usage patterns.

Pandas

Pandas was used for data manipulation and cleaning. It allows efficient handling of structured datasets and provides functions for filtering, grouping, and transforming data.

NumPy

NumPy was used for numerical computations and handling arrays during data preprocessing.

Matplotlib and Seaborn

These libraries were used for data visualization. Various plots such as bar charts, box plots, and heatmaps were created to analyze customer behavior.

Scikit-Learn

Scikit-Learn was used to implement the machine learning model. Logistic Regression was applied to predict whether a customer will churn.

Power BI

Power BI was used to create an interactive dashboard that displays key insights such as churn rate, device usage, and customer engagement.

Visual Studio Code

VS Code was used as the development environment to write and execute Python scripts.

5. Data Description

The dataset used in this project contains customer information from a streaming platform. Each row represents a unique customer and includes multiple attributes describing customer behavior and subscription details.

Important features in the dataset include:

Customer ID

A unique identifier assigned to each customer.

Subscription Type

Indicates the subscription plan chosen by the customer, such as Basic, Standard, or Premium.

Region

Represents the geographic location of the customer.

Device

Shows the device used by the customer to access the streaming platform.

Watch Hours

Represents the total number of hours a customer spends watching content.

Last Login Days

Indicates how many days have passed since the customer last logged into the platform.

Churned

This is the target variable used in the predictive model.

- 1 indicates the customer has churned.
- 0 indicates the customer is still active.

6. System Architecture

The system architecture of the customer churn prediction project consists of several stages that work together to process data and generate insights.

Data Source

Customer data containing subscription details, watch hours, device usage, and churn status is used as the input dataset.

Data Storage

The dataset is stored in a SQL database where queries are executed to analyze churn patterns.

Data Processing

Python libraries such as Pandas and NumPy are used for data cleaning and preprocessing.

Exploratory Data Analysis

Visualization libraries such as Matplotlib and Seaborn help identify patterns in customer behavior.

Machine Learning Model

A Logistic Regression algorithm is used to predict churn probability.

Model Evaluation

The model is evaluated using accuracy score, confusion matrix, and classification metrics.

Visualization Layer

Power BI dashboards present the results in an interactive and visual format.

7. Project Methodology

The project follows a structured data analytics workflow.

Data Collection

The dataset containing customer information was collected from and used as the primary data source.

Data Cleaning

Duplicate records were removed and missing values were handled to improve data quality.

Data Preprocessing

Categorical variables were encoded and column names were standardized.

Exploratory Data Analysis

Visualizations were created to analyze churn patterns and customer behavior.

Feature Engineering

Relevant features were selected and transformed to improve model performance.

Predictive Modeling

A Logistic Regression model was trained to predict customer churn.

Model Evaluation

The performance of the model was evaluated using various classification metrics.

Data Visualization

Power BI dashboards were developed to present insights interactively.

8. Data Cleaning and Preprocessing

Data preprocessing is an important step in the analytics process. It ensures that the dataset is clean and suitable for analysis.

The following steps were performed:

- Removal of duplicate records
- Handling missing values
- Standardization of column names
- Encoding categorical variables
- Splitting the dataset into training and testing sets

These steps helped prepare the dataset for machine learning modeling.

9. Exploratory Data Analysis

Exploratory Data Analysis was conducted to understand customer behavior and identify patterns that influence churn.

Several visualizations were created, including:

- Customer churn distribution
- Churn by subscription type
- Watch hours vs churn

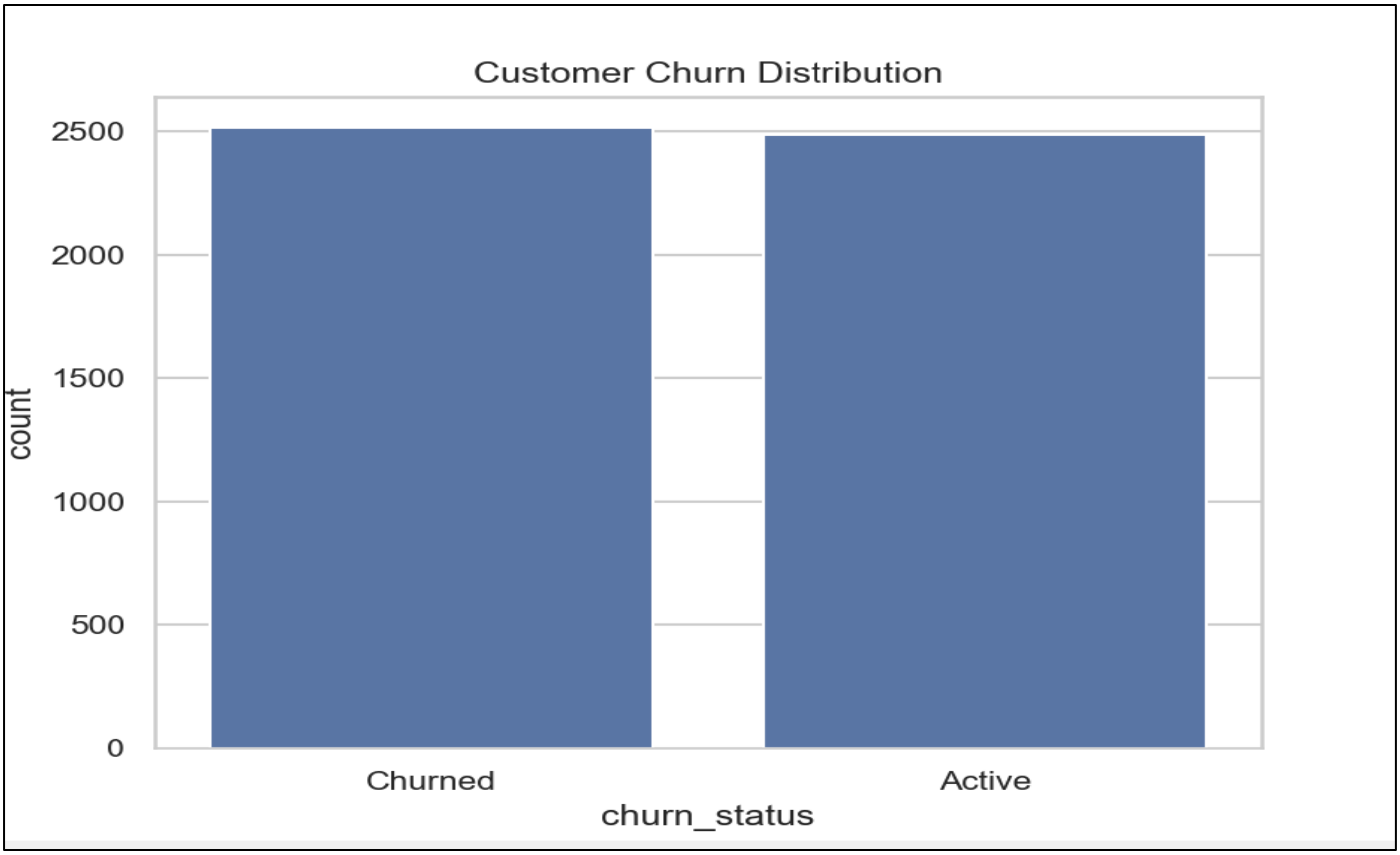
- login days vs churn
- Churn by region
- Correlation heatmap

These visualizations helped identify relationships between different features and churn behavior.

Plot 1 – Customer Churn Distribution

Explanation

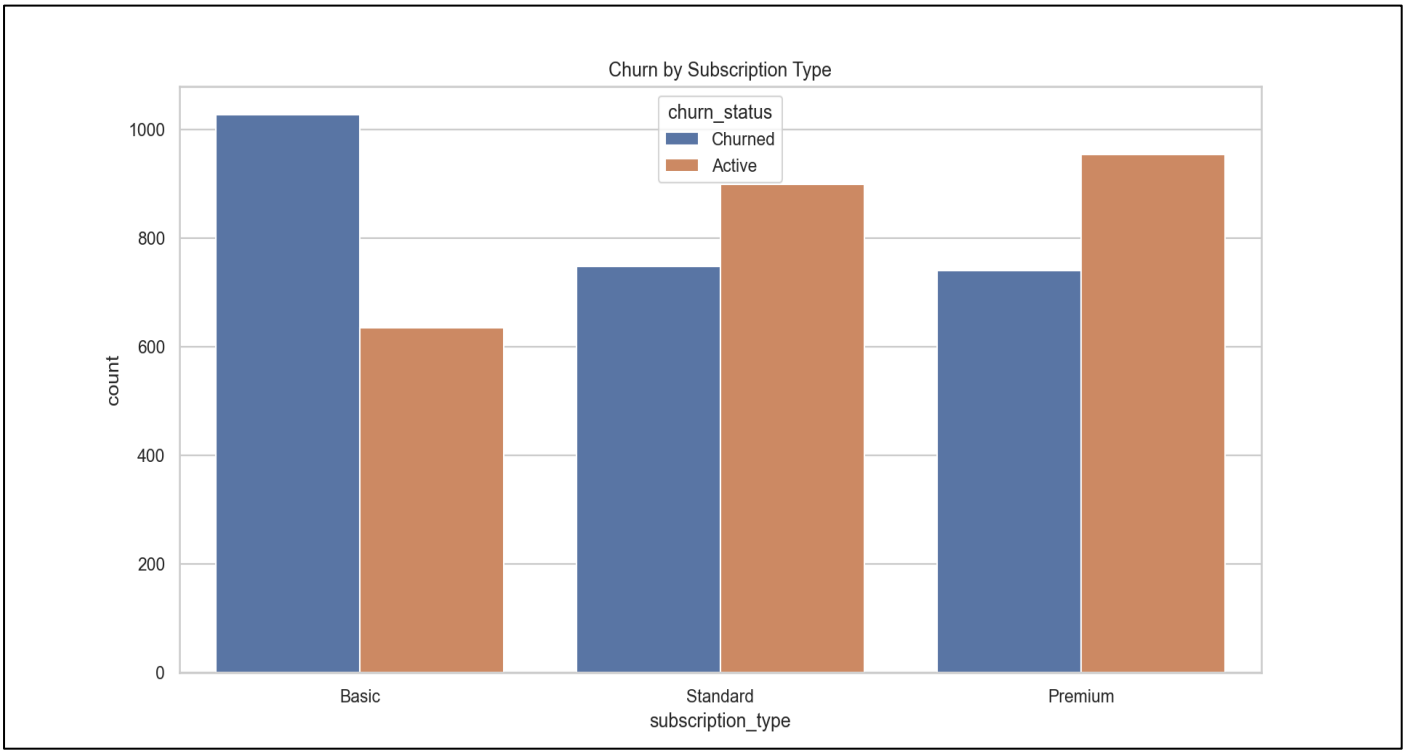
The churn distribution plot shows the proportion of customers who have remained active versus those who have discontinued their subscription. This visualization helps understand the overall churn rate in the dataset and provides an initial overview of customer retention levels.



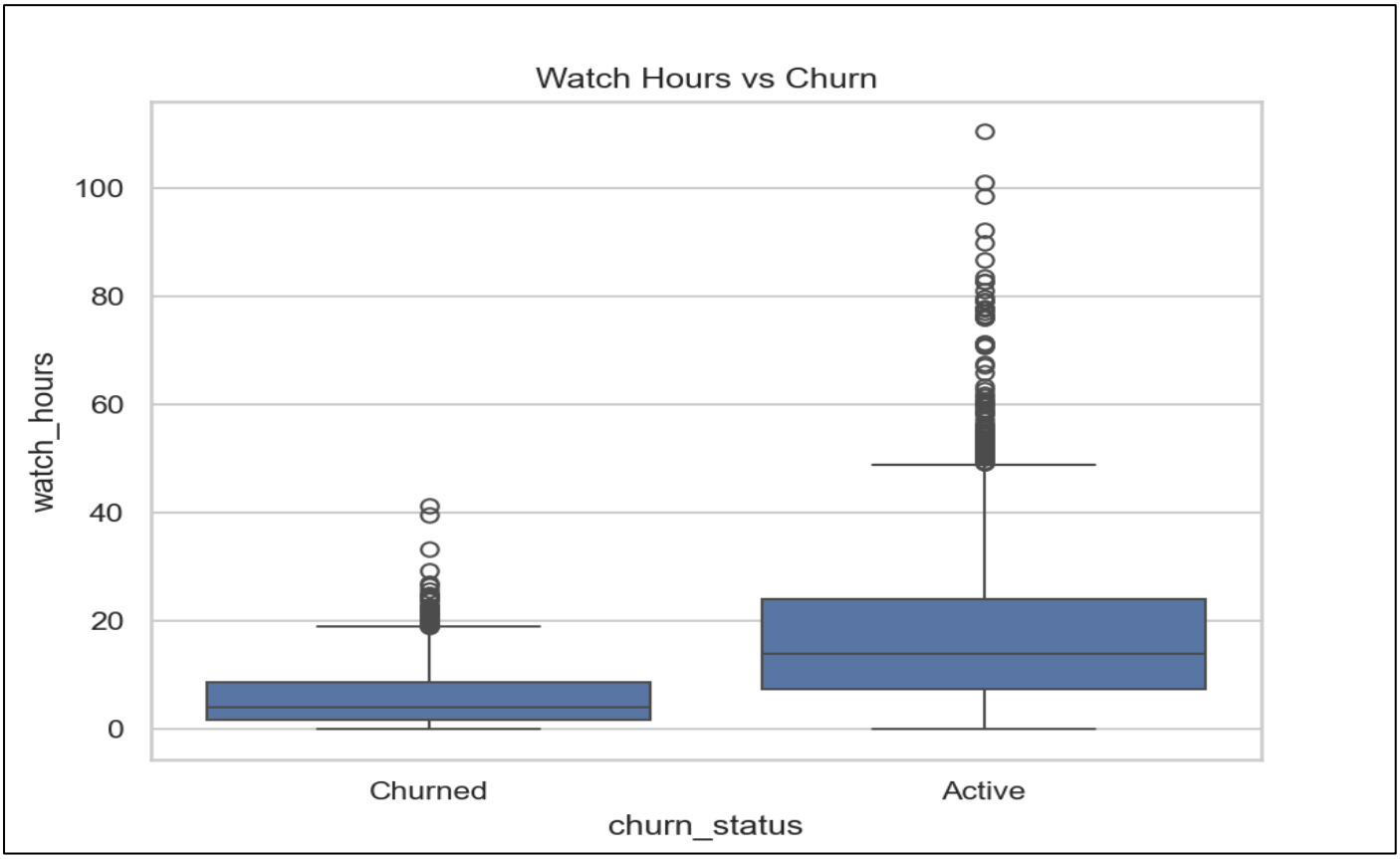
Plot 2 – Churn by Subscription Type

Explanation:

This plot shows churn behavior across different subscription plans. It helps identify whether certain subscription tiers experience higher churn rates compared to others.



Plot 3 – Watch Hours vs Churn



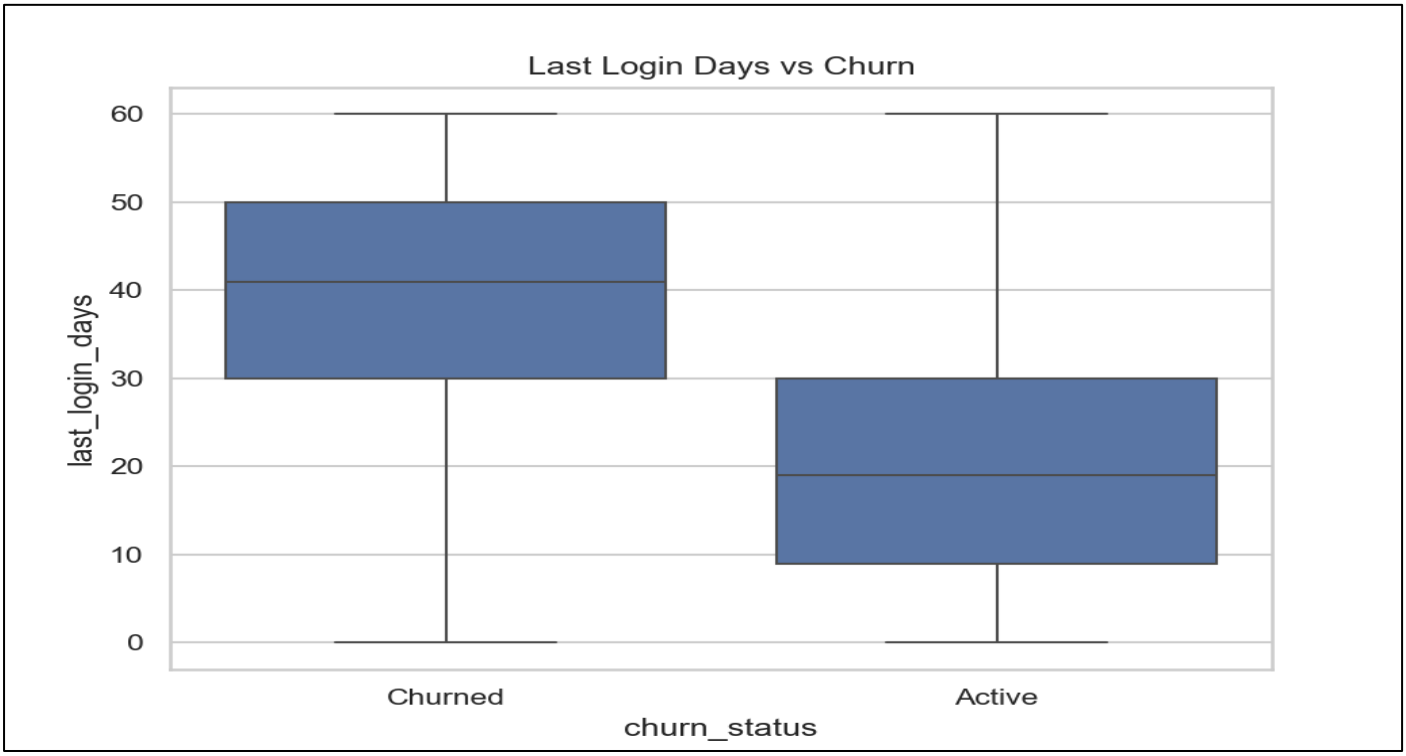
Explanation:

The box plot compares watch hours between active and churned customers. The analysis indicates that customers who spend more time watching content are less likely to churn.

Plot 4 – Last Login Days vs Churn

Explanation:

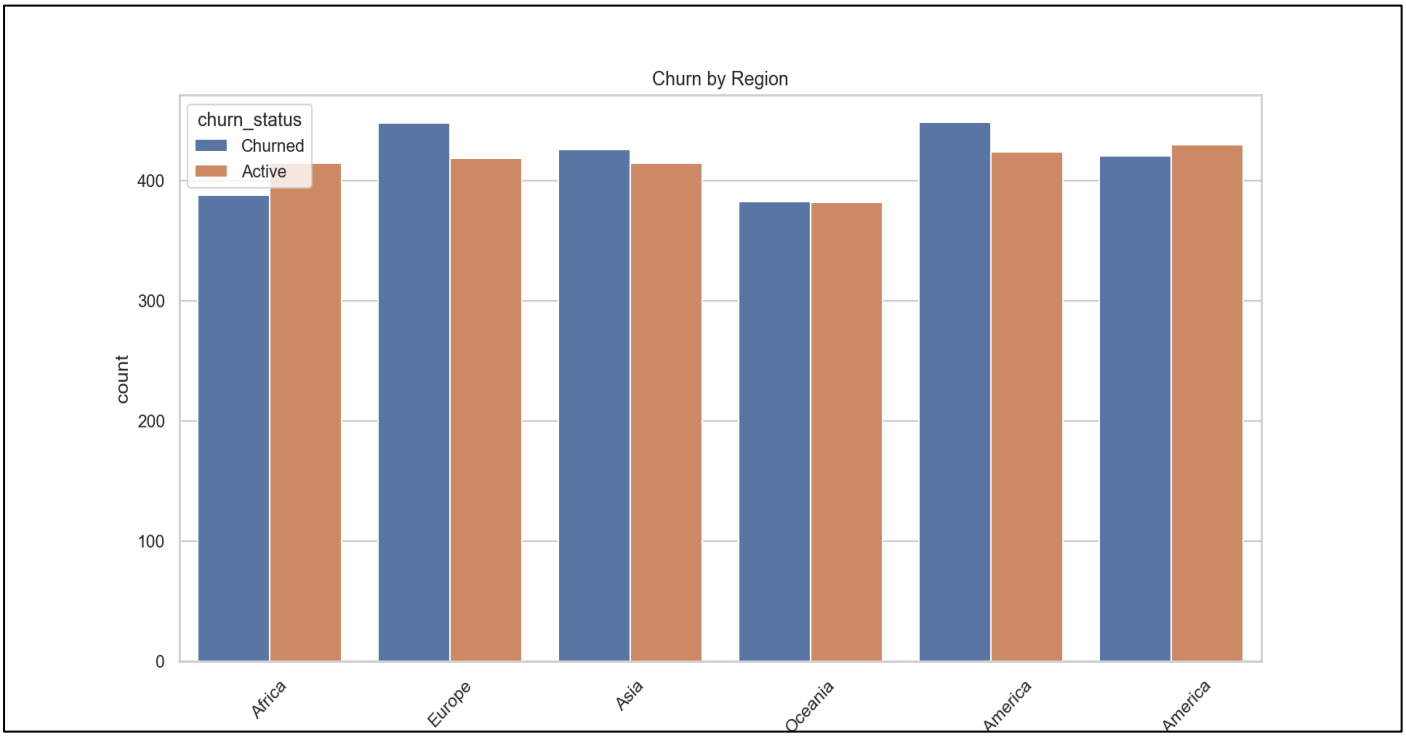
This visualization compares the average number of days since the last login for active and churned customers. Customers who have not logged in for a longer period show a higher likelihood of churn.



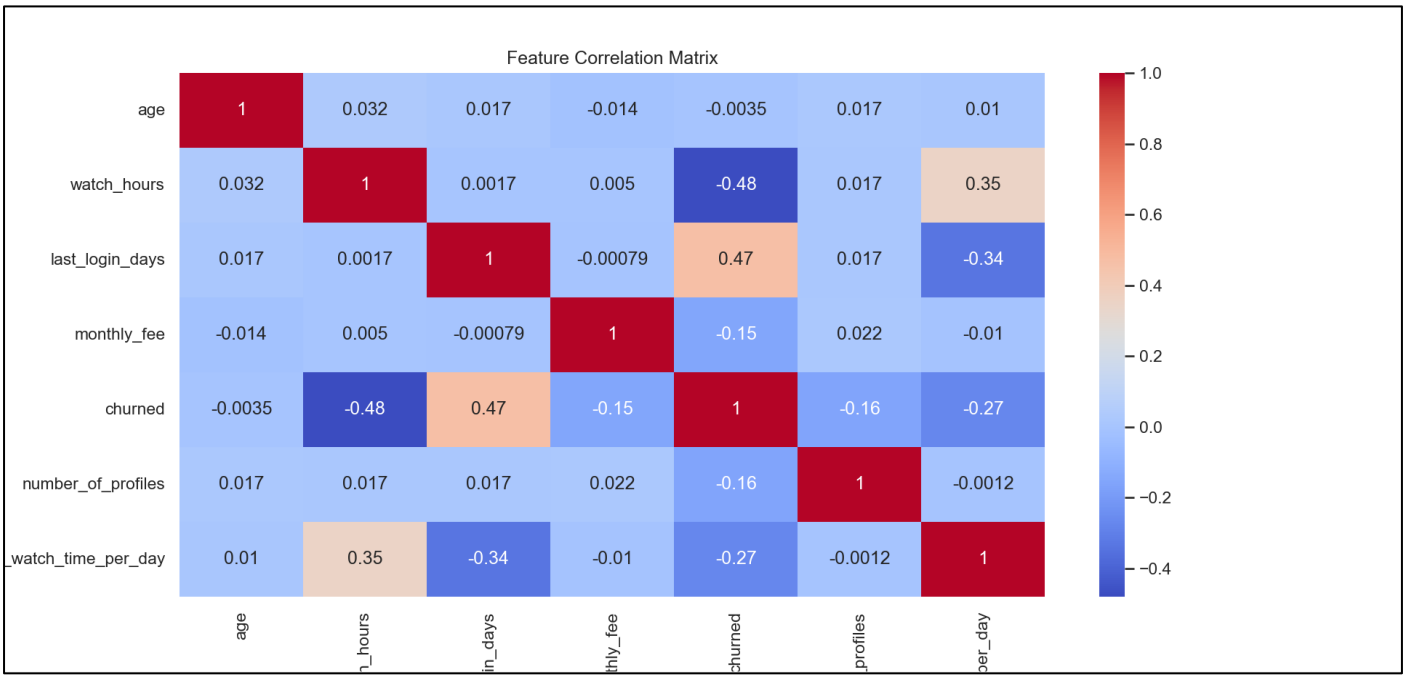
Plot 5 – Churn by Region

Explanation:

The regional churn plot shows how churn behavior varies across different geographic locations. This analysis helps identify regions where customer retention strategies may need improvement.



Plot 6 – Correlation Heatmap



Explanation:

The correlation heatmap displays relationships between numerical variables in the dataset. It helps identify which features are strongly related to churn and may influence the predictive model.

10. Feature Engineering

Feature engineering was performed to improve the predictive capability of the machine learning model. A new categorical feature called **churn status** was created to represent customer activity as “Active” or “Churned”. Additionally, categorical variables such as **subscription type, device, region, gender, and favorite genre** were converted into numerical values using **Label Encoding** so that the machine learning algorithm could process them effectively.

Furthermore, a **churn probability feature** was generated using the trained Logistic Regression model. This feature estimates the likelihood of each customer leaving the platform and helps identify high-risk customers.

11. Predictive Modeling

Logistic Regression was used as the machine learning algorithm to predict customer churn.

Logistic Regression is a supervised learning algorithm commonly used for binary classification problems.

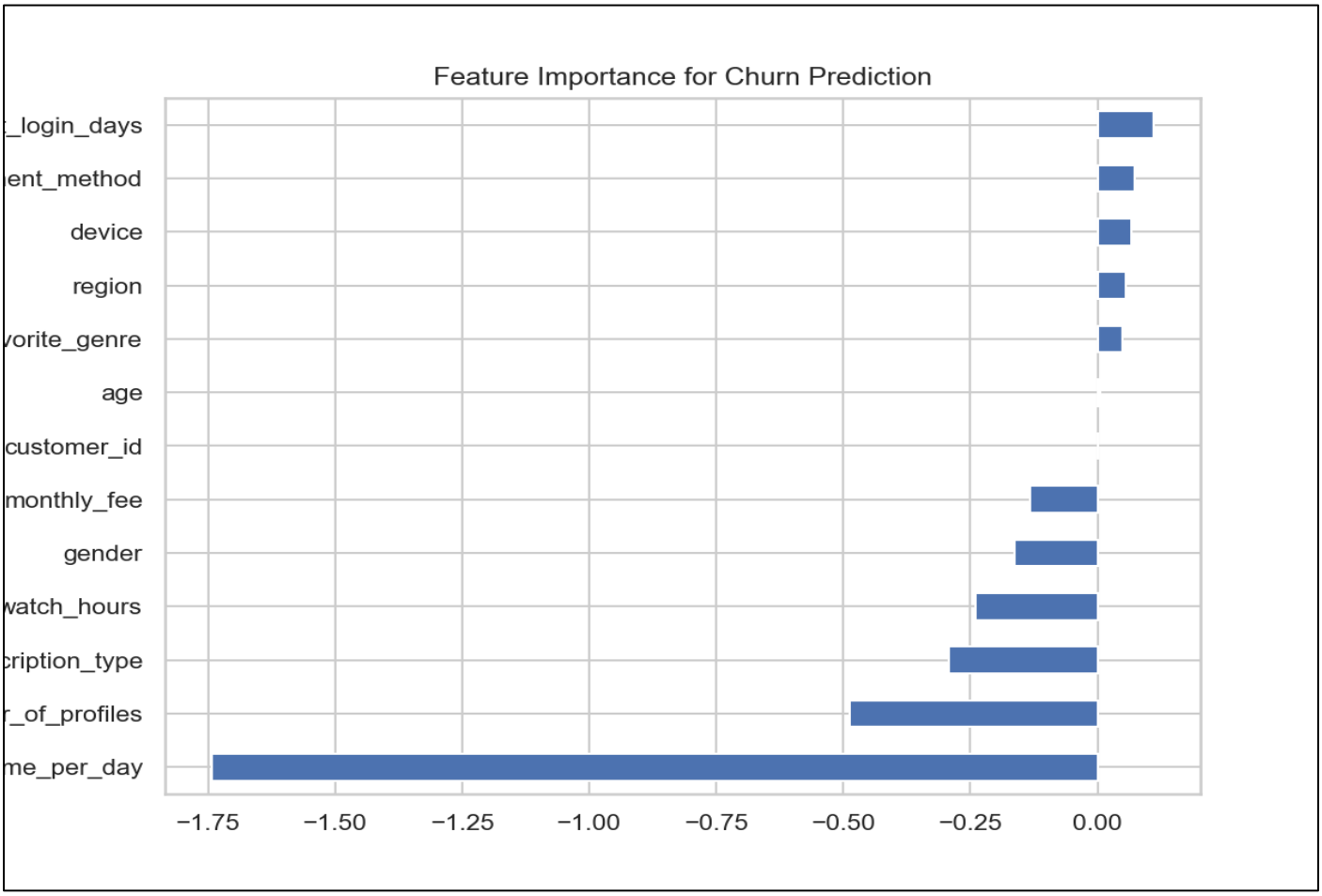
The dataset was divided into:

Training Data – 80%
Testing Data – 20%

The model was trained on the training dataset and then used to predict churn on the test dataset.

Plot 7 – Feature Importance Plot

Explanation:
The feature importance plot illustrates the variables that have the greatest influence on the churn prediction model. Features with higher importance scores contribute more significantly to predicting whether a customer is likely to churn. This visualization helps identify the key factors affecting customer behaviour and enables businesses to focus on the most impactful variables when developing strategies to improve customer retention.



12. Model Evaluation

The performance of the machine learning model was evaluated using several metrics.

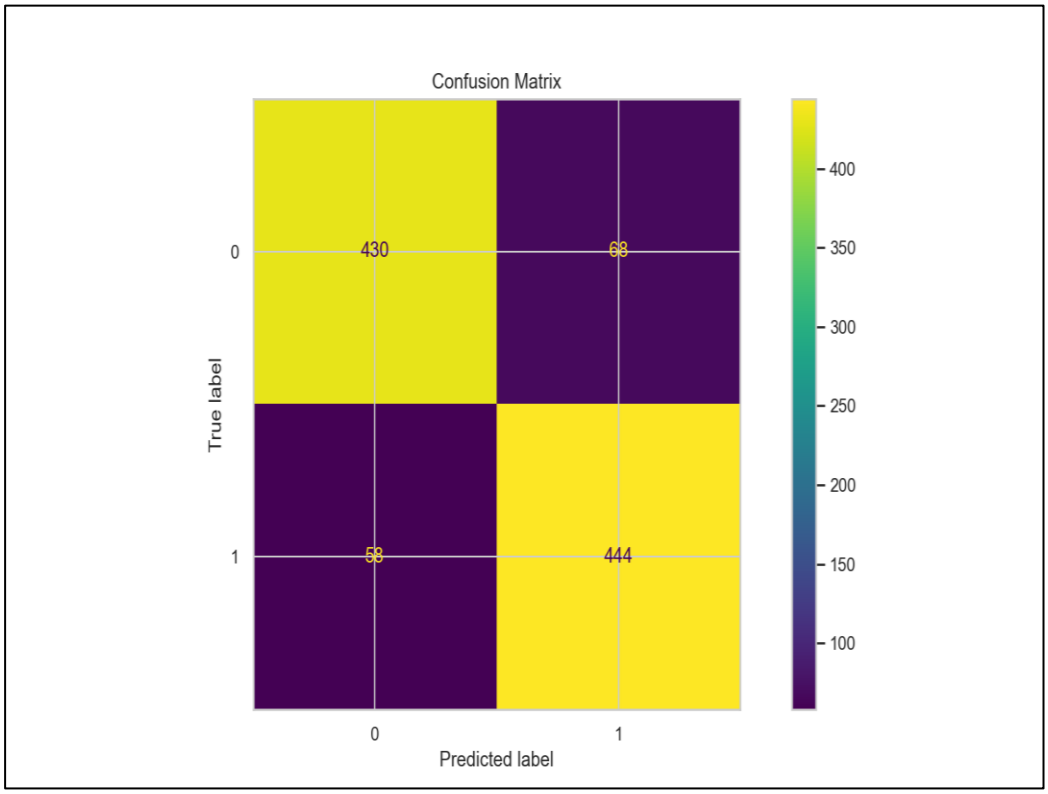
Accuracy

Accuracy measures the percentage of correct predictions made by the model.

Confusion Matrix – Explanation

The confusion matrix evaluates the performance of the churn prediction model by comparing the **actual customer status** with the **predicted results**.

- **True Negatives (430):** Customers correctly predicted as active.
- **False Positives (68):** Active customers incorrectly predicted as churn.
- **False Negatives (58):** Churned customers incorrectly predicted as active.
- **True Positives (444):** Customers correctly predicted as churn.



These results show that the model correctly identifies most churned and active customers, indicating good predictive performance.

13. Dashboard Interpretation

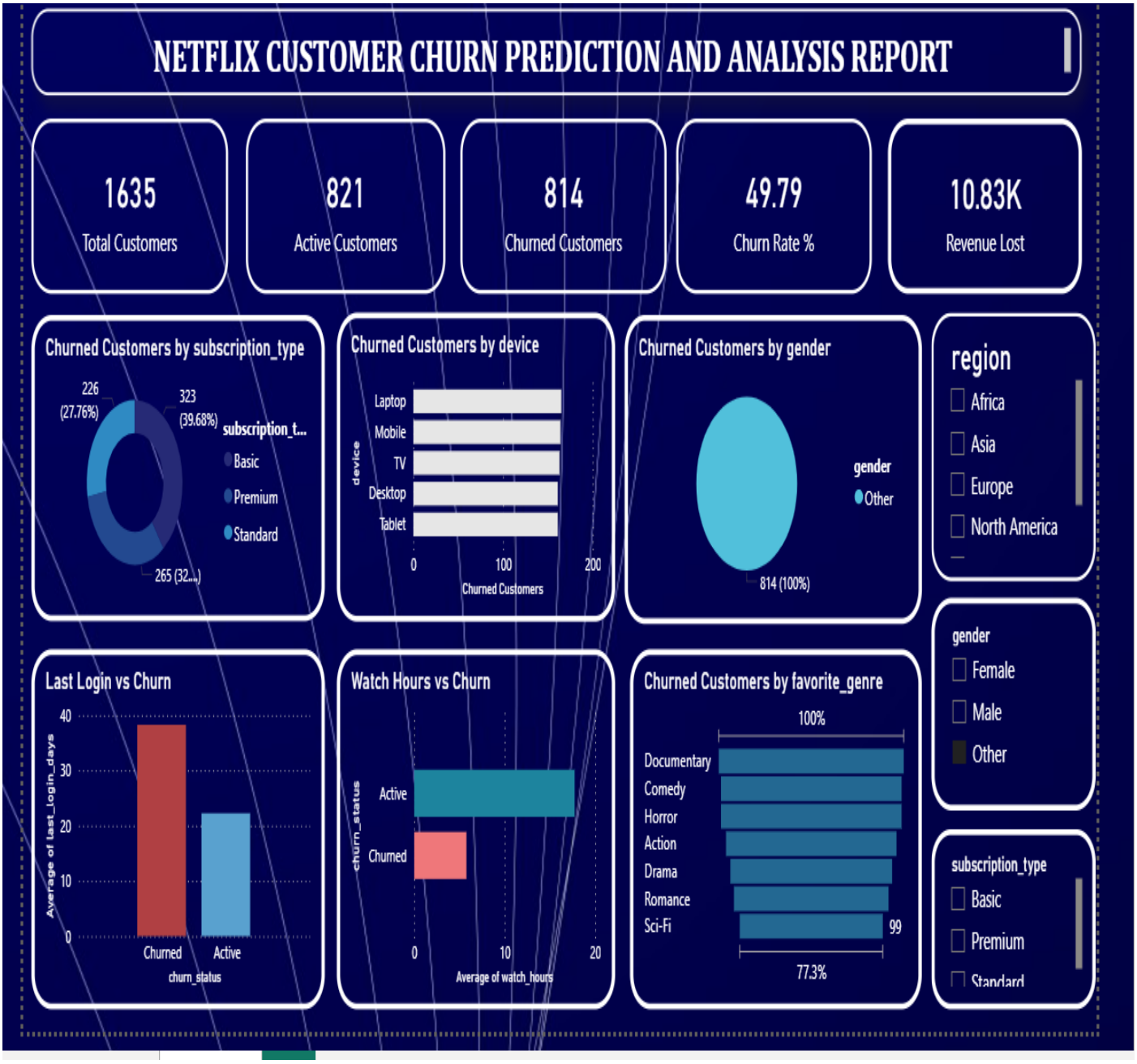
The Power BI dashboard provides a visual representation of customer churn patterns and key engagement metrics. It allows stakeholders to analyze customer behavior and identify factors that contribute to customer attrition.

Key Performance Indicators

The dashboard displays the following KPIs:

- Total Customers: 5000
- Active Customers: 2485
- Churned Customers: 2515
- Churn Rate: 50.30%
- Revenue lost: 10.83K

The churn rate indicates that approximately half of the customers have discontinued their subscription. This highlights the importance of identifying churn drivers and implementing effective retention strategies.



Total Customers

This KPI shows the total number of customers present in the dataset. It represents the overall customer base used to analyze churn behavior.

Active Customers

This card displays the number of customers who are currently using the platform. A higher active customer count indicates stronger engagement and better customer retention.

Churned Customers

This KPI represents the total number of customers who have stopped using the service. A high churn count indicates potential issues in customer satisfaction or engagement.

Churn Rate %

The churn rate shows the percentage of customers who have cancelled their subscription. A churn rate close to 50% indicates a significant retention challenge for the platform.

Revenue Lost

This KPI highlights the estimated revenue lost due to churned customers. It shows the financial impact of customer attrition on the business.

Customer Churn Distribution

The donut chart shows the proportion of active and churned customers. It helps quickly understand the overall churn situation within the customer base.

Churn by Subscription Type

This chart shows how churned customers are distributed across Basic, Standard, and Premium plans. It helps identify which subscription tier experiences the highest churn.

Churn by Device

The device chart displays churned customers based on the device used for streaming. This helps detect if certain devices experience higher churn due to usability or performance issues.

Churned Customers by Gender

This visualization shows the distribution of churned customers across different gender categories. It helps analyze whether demographic factors influence churn behavior.

Watch Hours vs Churn

This chart compares the average watch time between active and churned customers. It indicates that customers who spend more time watching content are less likely to churn.

Last Login vs Churn

This visualization compares the inactivity period between active and churned users. It shows that customers with longer inactivity periods are more likely to leave the platform.

Churned Customers by Favorite Genre

This chart shows churn distribution based on customers' preferred content genres. It helps understand whether content preference influences customer engagement and retention.

Region Filter

The region filter allows users to analyze churn patterns across different geographic locations. It helps identify regional differences in customer behavior.

Gender Filter

This interactive filter allows the dashboard to display churn insights based on selected gender categories. It helps in segmenting and analyzing churn trends across demographics.

Subscription Type Filter

The subscription filter allows users to analyze churn behavior for Basic, Standard, and Premium plans separately. It helps identify which plan requires retention improvements.

14. Business Insights

The analysis of customer behavior revealed several key insights that influence churn patterns and customer engagement.

• **Lower Engagement:** Customers with fewer watch hours are more likely to churn, indicating that reduced platform engagement is a strong churn indicator.

• **Login Activity:** Customers who have not logged in recently show a higher probability of churn, suggesting inactivity as an early warning sign.

• **Subscription Plans:** Certain subscription plans experience higher churn rates, indicating possible differences in perceived value or pricing.

• **Device Usage:** Device usage patterns affect customer engagement, as users interact differently depending on the device they use.

• **Regional Variation:** Differences across regions indicate varying customer retention levels, highlighting the need for region-specific strategies.

15. Business Recommendations

Based on the findings from the analysis, several strategies can be implemented to reduce customer churn and improve overall customer retention.

• **Personalized Content Recommendations:** Providing tailored content suggestions based on customer preferences can increase user engagement and encourage continued platform usage.

• **Engagement Reminders:** Sending notifications or reminders to inactive users can help re-engage customers who have reduced their activity on the platform.

• **Promotional Offers and Discounts:** Offering special discounts, trial extensions, or promotional subscription plans to low-engagement customers can motivate them to remain subscribed.

• **Improved User Experience:** Enhancing the platform's performance and usability across various devices can improve customer satisfaction and overall engagement.

• **Customer Loyalty Programs:** Implementing reward or loyalty programs for long-term customers can strengthen relationships and encourage continued service usage.

16. Conclusion

Customer churn prediction is a critical application of data analytics in subscription-based industries such as streaming platforms, telecommunications, and digital services. Understanding the factors that contribute to customer churn enables organizations to identify at-risk customers and implement proactive retention strategies.

This project demonstrated a comprehensive data analytics pipeline that included data preprocessing, exploratory data analysis (EDA), machine learning modeling, and dashboard visualization. Data preprocessing ensured the dataset was clean and reliable by handling missing values, removing duplicates, and standardizing formats. Exploratory analysis helped uncover patterns and relationships within the data, allowing a deeper understanding of customer behavior and engagement trends.

The results of the analysis indicate that engagement metrics such as watch hours, login frequency, and user activity levels have a significant influence on customer churn behavior. Customers with lower engagement levels were found to have a higher probability of discontinuing the service, highlighting the importance of maintaining consistent user interaction with the platform.

Furthermore, the integration of machine learning models allowed the identification of key predictive factors that contribute to churn risk. The use of visualization tools and interactive dashboards provided clear insights into customer segments, usage patterns, and churn indicators, enabling stakeholders to interpret the results effectively.

By leveraging predictive analytics and data visualization tools, businesses can make informed decisions and design targeted strategies to improve customer retention. Implementing personalized recommendations,

engagement campaigns, and customer loyalty initiatives can significantly reduce churn rates and enhance overall customer satisfaction.

Overall, this project demonstrates how data-driven approaches can support strategic decision-making and help organizations build sustainable customer relationships in competitive markets.

17. Future Scope

This project can be further improved and expanded by incorporating more advanced analytical techniques and real-world implementations.

• **Advanced Machine Learning Models:** Future work can include implementing advanced algorithms such as Random Forest and XGBoost to improve the accuracy of churn prediction.

• **Real-Time Prediction Systems:** Developing real-time churn prediction systems can help businesses identify at-risk customers and take preventive actions immediately.

• **Recommendation Systems:** Integrating recommendation systems can increase customer engagement by providing personalized content and product suggestions.

• **Additional Data Sources:** Including more data such as detailed viewing history, customer feedback, and behavioral data can enhance the quality of analysis and predictions.

• **Model Deployment:** The predictive model can be deployed as a web application or integrated into business systems to support real-time decision-making in practical scenarios.

18. References

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